

An Exact Three-Disk Certificate for Packing Disks of Radii 1, 2, 3, 4, 5

Proof note — for independent review

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Theorem 1. *The minimum radius of a circular container that holds five pairwise interior-disjoint disks of radii 1, 2, 3, 4, 5 is*

$$R_* = \frac{2820 + 1440\sqrt{5}}{671}, \quad 671R_*^2 - 5640R_* - 3600 = 0, \quad 9 < R_* < \frac{901}{100}.$$

Lemma 1 (Three directions). *For any three nonzero vectors in the plane, the sum of the smaller angles between their three pairs is at most 2π .*

Proof. List the three directions cyclically and let their successive gaps be $g_1, g_2, g_3 \geq 0$ with sum 2π . If all gaps are at most π , the three smaller pair angles are those gaps and sum to 2π . Otherwise precisely one gap, say g_3 , exceeds π . The smaller pair angles are then $g_1, g_2, 2\pi - g_3 = g_1 + g_2$, whose sum is $2(g_1 + g_2) < 2\pi$. $\square$

Proof of the theorem. 1. *A universal lower bound from the disks of radii 3, 4, 5.* Write p_i for the disk centers and $s_i = \|p_i\|$. Two disks of radii 4, 5 force $R \geq 9$. To prove the lower bound, we may suppose $9 \leq R \leq R_*$, since a larger container already satisfies it. In particular, $R < 9.01$.

Containment gives $s_i \leq R - i$ and separation plus the triangle inequality gives $s_i + s_j \geq i + j$ for distinct i, j in $\{3, 4, 5\}$. In particular $s_3 \geq 13 - R > 0$, $s_4 \geq 14 - R > 0$, and $s_5 \geq 13 - R > 0$; all relevant center directions are defined.

For a pair i, j , put $d = i + j$ and

$$c(x, y) = \frac{x^2 + y^2 - d^2}{2xy}, \quad \phi_{ij}(x, y) = \arccos c(x, y).$$

For all radial values arising here, $0 < x \leq R - i < d$ and $0 < y \leq R - j < d$. Moreover $x + y \geq d$ and $|x - y| < d$. The radial cosine satisfies

$$\frac{\partial c}{\partial x} = \frac{x^2 - y^2 + d^2}{2x^2y} > 0, \quad \frac{\partial c}{\partial y} = \frac{y^2 - x^2 + d^2}{2xy^2} > 0.$$

Increasing each radial coordinate to its wall value $R - i$, $R - j$ therefore can only decrease the necessary smaller angle. Consequently, for the actual smaller pair angle γ_{ij} ,

$$\gamma_{ij} \geq \phi_{ij}(s_i, s_j) \geq \beta_{ij}(R), \quad \beta_{ij}(R) = 2 \arcsin \sqrt{\frac{ij}{(R - i)(R - j)}}.$$

This remains valid at the limiting value $x + y = d$ by continuity. Applying the three-directions lemma yields

$$\beta_{34}(R) + \beta_{35}(R) + \beta_{45}(R) \leq \gamma_{34} + \gamma_{35} + \gamma_{45} \leq 2\pi. \quad (1)$$

Each β_{ij} strictly decreases as R increases, for $R > 9$.

2. *Exact construction at R_* .* Set $R = R_*$ and $A = R - 4$. Define

$$\begin{aligned} x_3 &= \frac{R^2 - 7R - 12}{R - 4}, & u &= \frac{48R(R - 7)}{(R - 4)^2}, \\ x_5 &= \frac{R^2 - 9R - 20}{R - 4}, & v &= \frac{80R(R - 9)}{(R - 4)^2}. \end{aligned}$$

Place

$$p_4 = (A, 0), \quad p_3 = (x_3, \sqrt{u}), \quad p_5 = (x_5, -\sqrt{v}).$$

Direct algebra gives

$$\begin{aligned} x_3^2 + u &= (R - 3)^2, & x_5^2 + v &= (R - 5)^2, \\ (x_3 - A)^2 + u &= 7^2, & (x_5 - A)^2 + v &= 9^2. \end{aligned}$$

Thus all three disks touch the container; disks 3, 4 and 4, 5 touch each other. For the remaining pair, let

$$h = \frac{64 - (x_3 - x_5)^2 - u - v}{2} = -\frac{2(17R^2 - 128R - 240)}{(R - 4)^2}.$$

A direct polynomial identity gives

$$h^2 - uv = -\frac{4(671R^2 - 5640R - 3600)}{(R - 4)^2} = 0.$$

Since $9 < R < 9.01$ implies $h > 0$, we have $h = \sqrt{uv}$, and hence

$$\|p_3 - p_5\|^2 = (x_3 - x_5)^2 + (\sqrt{u} + \sqrt{v})^2 = 64.$$

Therefore the three large disks are mutually tangent.

The same rational bounds $9 < R < 9.01$ give

$$0 < x_3 < 2, \quad \sqrt{u} > 5, \quad x_5 < -\frac{39}{10}, \quad -1 < -\sqrt{v} < 0.$$

In particular p_4 lies on the positive horizontal ray, p_3 lies in the first quadrant, and p_5 lies in the third quadrant. Their cyclic gaps $4 \rightarrow 3$, $3 \rightarrow 5$, $5 \rightarrow 4$ are all smaller than π : for the middle gap, the determinant satisfies

$$\det(p_3, p_5) = x_3(-\sqrt{v}) - (\sqrt{u})x_5 > -2 + 5 \cdot \frac{39}{10} > 0.$$

Consequently the three necessary wall angles sum to exactly 2π :

$$\beta_{34}(R_*) + \beta_{35}(R_*) + \beta_{45}(R_*) = 2\pi.$$

The strict decrease of the left side of (1) therefore rules out every $9 \leq R < R_*$. We have proved that any packing of these three disks, hence of all five disks, needs $R \geq R_*$.

3. *Insert the two smaller disks.* Place the remaining disk centers at the *rational* points

$$p_1 = (5, -6), \quad p_2 = (1, -6).$$

They lie inside the container since $\|p_1\|^2 = 61 < 8^2 < (R - 1)^2$ and $\|p_2\|^2 = 37 < 7^2 < (R - 2)^2$. The two centers are distance $4 > 3$ apart. Both lie at least 6 vertically below disk 3, whose ordinate exceeds 5; thus they are separated from it. Their distances from disk 4 exceed 5 and 6, respectively, the first by its vertical gap and the second by its nonzero horizontal gap. Finally, $x_5 < -3.9$ shows $\|p_1 - p_5\| > 8.9 > 6$, and $-1 < y_5 < 0$ gives

$$\|p_2 - p_5\|^2 > (4.9)^2 + 5^2 = 49.01 > 7^2.$$

Thus all ten separation constraints and all five containment constraints hold at $R = R_*$. This supplies the matching upper bound and finishes the proof. $\square$